WORKBOOK 03 / 08 · MULTIMEDIA 12

Make the Caveman think, then point

Key poses, designed breakdowns, in-betweens, timing and spacing

THE BREAKDOWN DESIGNS THE ROUTE



13 · thought



19 · hand clears beard



25 · point

A designed breakdown clears the hand around the face and lets the head lead.

Build: Build a short acting phrase whose transition is designed rather than left to default interpolation.

Keep: Think_Point Graphic + stepped and refined versions + a timing comparison

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

Keys state the important poses. A breakdown designs how the action travels between them. In-betweens complete that designed transition.

A timeline keyframe stores a change; a storytelling key pose is an important idea in the performance. They are not synonyms. Timing is how long an action takes. Spacing is how positions are distributed through that time. A classic tween interpolates transforms; it does not understand anatomy, contact or intention.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

The Caveman notices a distant noise before pointing at the stone. Add a head-led change of attention without losing the original story goal.

What you will defend

Which drawing is your breakdown, and what does it decide that a mathematical midpoint would miss?

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Block the two storytelling keys

SELECT: A new Think_Point Graphic copied from the neutral character.

TOOLS / ROUTE: Library → Duplicate; symbol timeline; Insert Keyframe.

1. Keep neutral at frame 1. At frame 13 pose a clear thought: head inclines toward the beard hand. At frame 25 pose a clear point toward the goal.

2. Extend every part layer through frame 25. Keep only held keys while judging the performance; do not tween yet.

3. Write the intention under each key: “I am considering” and “There is my goal.”



Storytelling keys communicate consideration and a decision.

WHY · Keys must communicate the action before smoothness can improve it.

CHECK / SUMMARY · The two important poses have distinct silhouettes and the pointing hand indicates the goal.

FIX · If the point reads as a wave, straighten the intended arm direction and make the head/gaze support it.

GLOSSARY · key pose: An important pose that communicates the action or intention.; stepped blocking: Reviewing held poses before creating transitions.

DIFFERENT-OBJECT EXAMPLE · For a cat turning to a sound, the starting gaze and final gaze are keys; they do not yet specify the route.

STRETCH · Make the thought register using a torso shift as well as a hand position.

02 · Design the breakdown at frame 19

SELECT: All animated part layers at frame 19 inside Think_Point.

TOOLS / ROUTE: Insert Keyframe; Onion Skin; Free Transform.

1. Insert keys at frame 19 between the thought and point. Move the elbow outward so the hand clears the beard before extending.

2. Let the head begin to face the goal before the pointing arm fully extends. Keep shoulder, elbow and wrist connected.

3. Compare this pose with a simple halfway blend. Label the decision: “hand clears face; head leads.”

WHY · A breakdown chooses an arc, body relationship and order of movement. It need not be halfway in space or halfway in time.

CHECK / SUMMARY · The hand passes around the face rather than through it, and the viewer can follow the intended route.

FIX · If the hand cuts through the beard, widen the elbow path or move the breakdown earlier; do not add random frames around a bad route.

GLOSSARY · breakdown: A designed transition pose that explains how keys connect.; arc: A curved path followed by a moving point.

DIFFERENT-OBJECT EXAMPLE · A bird’s wing can fold near the body during recovery rather than taking the shortest straight-line route.

STRETCH · Place the breakdown at frame 17 instead and describe the change in emphasis.

03 · Add purposeful in-betweens

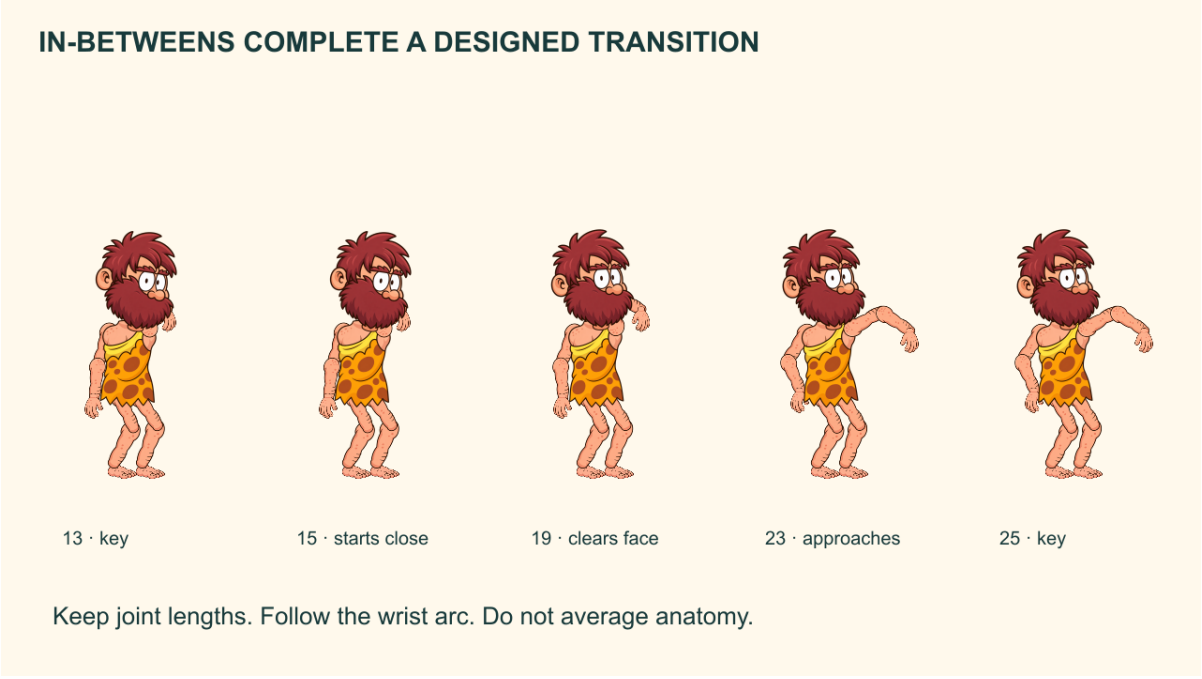
SELECT: Part layers between frames 13–19 and 19–25.

TOOLS / ROUTE: Insert Keyframe; Onion Skin; frame-by-frame posing.

1. Add in-between poses at frames 15, 17, 21 and 23. Keep the wrist on a smooth arc and maintain limb lengths.

2. Place the first movement close to the thought pose, allow larger spacing through the extension, then reduce spacing near the final point.

3. Play frames 13–25 repeatedly. Remove any in-between that introduces a kink; redraw or repose it instead of hiding it with more frames.



Transition planning strip. Additional poses follow a designed route while preserving joint lengths.

WHY · Additional drawings help only when their placement supports the intended motion.

CHECK / SUMMARY · The wrist path is smooth, joints remain connected and the final point does not snap unexpectedly.

FIX · If the forearm changes length, restore the correct elbow-to-wrist distance. If speed is uneven accidentally, compare adjacent wrist spacing.

GLOSSARY · in-between: A pose or drawing filling the transition between designed poses.; spacing: The distance between positions on successive frames.

DIFFERENT-OBJECT EXAMPLE · A pendulum has close positions near its turning points and wider positions through the bottom.

STRETCH · Try animating on twos, then add selected single-frame accents where the action needs precision.

04 · Compare timing while preserving the route

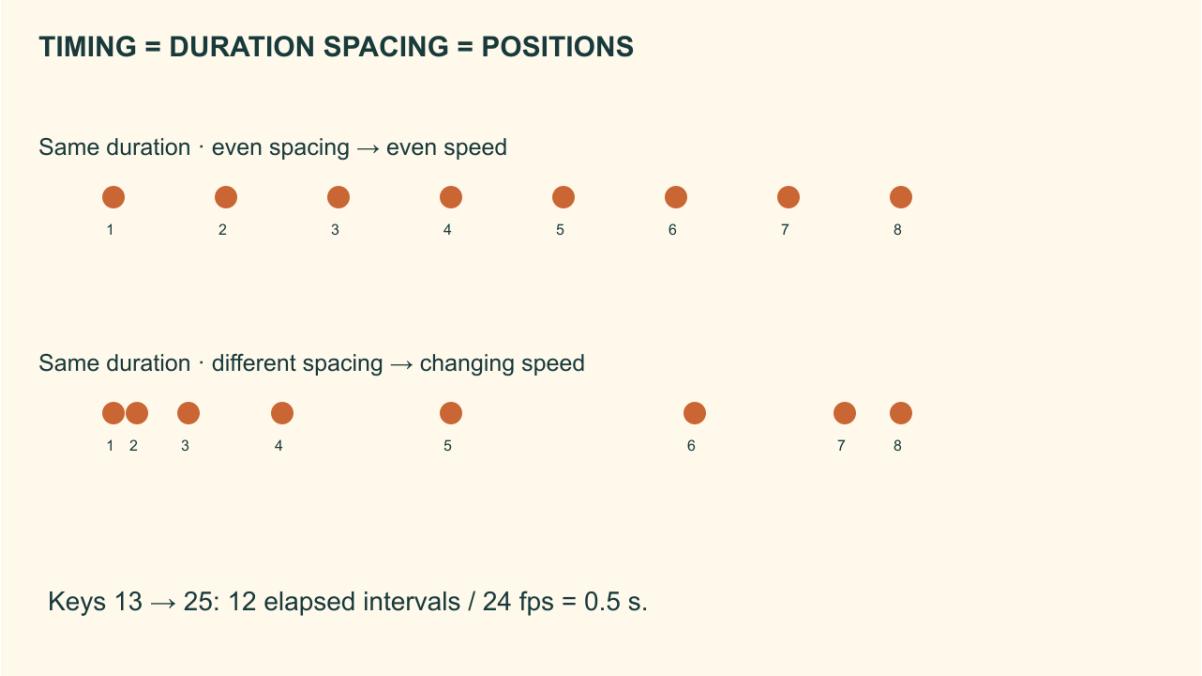
SELECT: A duplicate Think_Point_Slow; same poses and arc.

TOOLS / ROUTE: Timeline → move keyframes and extend all part layers.

1. Duplicate the action. In the slow version, move the final point from frame 25 to frame 37 and distribute the transition keys over the longer interval.

2. Keep the designed arc and relative pose order. Compare the two versions side by side or in sequence.

3. Calculate elapsed time between key 13 and key 25: 12 intervals ÷ 24 = 0.5 seconds. A 24-frame exported clip itself lasts one second.



Same frame intervals, different position spacing: duration and speed distribution are distinct.

WHY · Changing duration can change intention from quick recognition to careful consideration without changing the spatial route.

CHECK / SUMMARY · You distinguish interval duration from inclusive frame count and can explain the acting difference.

FIX · If a layer ends early, extend it to the new final frame. Do not change the document fps to make one action slower.

GLOSSARY · timing: The duration and placement of events in an action.; frame interval: The elapsed time from one frame position to the next.

DIFFERENT-OBJECT EXAMPLE · A lid closing over 6 intervals feels brisk; the same route over 24 intervals feels deliberate.

STRETCH · Keep duration fixed and change only spacing; describe why the result differs.

05 · Use a classic tween as a comparison

SELECT: A duplicate test action with one Graphic instance on each animated part layer.

TOOLS / ROUTE: Right-click between two keys → Create Classic Tween.

1. On the duplicate, test a classic tween between two pose keys. Select the span, not the Stage object, to inspect tween properties.

2. Scrub the midpoint and the elbow/hand contact. If a rotation takes the long way, inspect rotation settings and use the shortest suitable route.

3. Retain manual breakdown keys where the default interpolation produces a poor arc or broken joint. Do not claim the default tween designed your performance.

WHY · A tween can save repetitive placement, but it must be judged against pose and path decisions.

CHECK / SUMMARY · You can show an interpolated midpoint, identify a problem or justify why it works, and preserve your intended breakdown.

FIX · A dashed or failed span usually means unsuitable contents or mismatched symbols. Use one appropriate instance per layer and matching definitions at the ends.

GLOSSARY · classic tween: Interpolation of properties of an instance between keyframes.; interpolation: Calculating intermediate values between defined values.

DIFFERENT-OBJECT EXAMPLE · A sliding title is a good interpolation task; a hand avoiding a face needs an intentionally designed route.

STRETCH · Test uniform versus eased timing on the pointing hand and justify the selected version.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

06 · Add anticipation and a restrained settle

SELECT: Think_Point final pose and following frames.

TOOLS / ROUTE: Insert Keyframe at 11, 25, 28 and 32 as needed.

1. Extend EVERY part layer through frame 32 and extend the placed action's parent exposure to match. Before the point, let the hand draw slightly inward and the torso prepare. Keep this anticipation small enough to preserve the thought.

2. Let the arm reach a little beyond its settled point at frame 25, recover at 28 and rest at 32. Repose joints; do not merely wobble the whole picture.

3. Allow a smaller head or hand follow-through a frame or two after the torso settles.

WHY · Preparation makes the action legible; overshoot and settle suggest effort being absorbed.

CHECK / SUMMARY · The performance has a clear main action with a smaller recovery, rather than repeated bouncing.

FIX · If the settle distracts, reduce its amplitude and number of changes. Check the final pointing direction again.

GLOSSARY · anticipation: A preparation that helps viewers read the coming action.; overshoot: Passing slightly beyond the final position before returning.; settle: A decreasing recovery into rest.

DIFFERENT-OBJECT EXAMPLE · A person reaching for a shelf may shift weight first, reach beyond the target slightly and relax into the hold.

STRETCH · Make a cautious and an excited version using the same final pointing pose.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Diagnose the straight-line hand

SELECT: A fresh two-key comparison duplicate; preserve your designed version.

TOOLS / ROUTE: Delete selected transition keys only in the test copy.

1. In the comparison copy, keep only the thought and point endpoint keys for the transition. Remove the manual breakdown AND its manual in-betweens, then create classic spans with one matching Graphic instance on each animated part layer. Play this default interpolation and observe whether the hand cuts through the face or loses readability. If that particular pose pair does not cause an intersection, record the actual path/silhouette difference instead of inventing a fault.

2. Restore or redesign the breakdown and compare the wrist path with Onion Skin.

3. Explain the repair in terms of route and silhouette, not simply the number of frames.

WHY · The repair tests whether you understand what a breakdown contributes.

CHECK / SUMMARY · The repaired route has a visible clearance and retains the character's joint proportions.

FIX · If the route is still unclear, plot five wrist positions on a guide layer and redesign the curve before making more keys.

GLOSSARY · path of action: The route followed by a chosen part through the movement.

DIFFERENT-OBJECT EXAMPLE · A bouncing ball through a wall needs a different route, not just more in-between positions.

STRETCH · Make the repair with fewer transition poses while keeping the motion readable.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct the head-led reaction

SELECT: Your independent Think_Point variant.

TOOLS / ROUTE: New action duplicate; frame keys; full movie preview.

1. Add the changed-condition head turn before the pointing hand. Preserve the Caveman's eventual attention to the third stone.

2. Create keys, one labelled breakdown and at least two deliberate in-betweens. State which body part leads and which follows.

3. Save and reopen. Keep a pose strip, a timing note and your brief defence. Place the selected version in Discovery. Use a one-shot Graphic with First 1 and enough parent exposure; stop on a deliberate final pose instead of letting the phrase loop.

WHY · A new attention cue tests acting and sequence design rather than copying frame numbers.

CHECK / SUMMARY · The noise reaction is readable but does not replace the story's goal or create an unexplained extra event.

FIX · If the head and arm move as one stiff block, offset their starts and preserve a clear pause for recognition.

GLOSSARY · lead and follow: A deliberate order in which body parts start or finish movement.

DIFFERENT-OBJECT EXAMPLE · Independent task: use the same pose and timing criteria, without a supplied solution.

STRETCH · Use a subtle chest turn to support the head-led reaction.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

Original teacher-supplied Caveman PDF/PNG assets are preserved. New matching Caveman parts/backgrounds are AI-generated and identified in the asset register. Teaching diagrams are authored illustrations; genuine native captures are labelled separately. Supplied commercial course videos/FLAs are reference material and are not redistributed.

Publication/classroom status: see the teacher companion. This file does not certify the independent cold run, a natural voice performance, or live assessment/progress connections.